

LECTURE 4 Supplement

Topics

Holding returns for a portfolio with or w/o balancing, Optimal portfolio of one risky asset and the risk-free asset, Optimal risky portfolio of two risky assets, Complete portfolio of two risky assets and the risk-free asset

- Suppose you have \$1 million on Jan 31, 2000 and want to build a portfolio that allocates 20% to AAPL and 80% to IBM
- The data for AAPL and IBM is provided in the file datawork3.xls
- Suppose you want to hold the portfolio until June 30, 2022

Questions:

- 1 How much money would you have in your portfolio by June 30, 2022, if you were to rebalance your portfolio every month?
- 2 How much money would you have in your portfolio by June 30, 2022, if you were to create the portfolio on Jan 31, 2000 and then never rebalance your portfolio?

- A mutual fund, AA, has an expected return of 15% and a return volatility of 20%
- T-Bills rate is 4%
- An investor inherited a portfolio that currently allocates 80% to fund AA and the rest in T-Bills

Questions:

- 1 What is the expected return and the return volatility of the investor's portfolio?
- 2 Suppose the investor has a risk aversion coefficient of $A = 2.5$. What should be the optimal portfolio of T-Bills and fund AA for the investor?

- You are interested in investing in two risky assets, asset A1 and asset A2, and the T-Bills
- Consider the following data:

Asset	Expected return	Return volatility
A1	12%	25%
A2	9%	18%
T-Bills	4%	0%

- The covariance between the risky assets A1 and A2 is 0.03
- The covariance between the T-Bills and either A1 or A2 is 0

Questions:

- 1 What are the weights of the minimum variance portfolio of the risky assets A1 and A2?
- 2 What are the expected returns and return volatility of the minimum variance portfolio?
- 3 What are the expected return, return volatility and Sharpe ratio of a portfolio that allocates 70% to the minimum risky portfolio and the rest to the T-Bills?

- You are interested in investing in two risky assets, A1 and A2, and the T-Bills
- Consider the following data:

Asset	Expected return	Return volatility
A1	12%	25%
A2	9%	18%
T-Bills	4%	0%

- The covariance between the risky assets A1 and A2 is 0.03
- The covariance between the T-Bills and either A1 or A2 is 0

Questions:

- 1 What are the weights of the optimal risky portfolio, T, that consists of A1 and A2?
- 2 What are the expected return, return volatility and Sharpe ratio of a portfolio that allocates 70% to the optimal risky portfolio T and the rest to the T-Bills?
- 3 If your risk aversion is $A = 1.8$, what should be the weights of your optimal complete portfolio on the efficient frontier generated by A1, A2 and the T-Bills?
- 4 What should be the weight of the complete portfolio in stock A1?

- You are interested in investing in a portfolio that includes three risky assets, A1, A2, and A3
- Consider the following expected return and volatility data:

Asset	Expected return	Return volatility
A1	8%	17%
A2	9%	19%
A3	11%	18%
T-Bills	4%	0%

- And also the Covariance matrix:

Asset	A1	A2	A3
A1	2.89%	1.7%	-1.3%
A2	1.7%	3.61%	1.5%
A3	-1.3%	1.5%	3.24%

Questions:

- 1 What are the weights of the minimum variance portfolio consisting of A1, A2 and A3?
- 2 What are the weights of the optimal risky portfolio, T, consisting of A1, A2, and A3?
- 3 What are the expected return, return volatility and Sharpe ratio of a portfolio that allocates 110% to the optimal risky portfolio T and the rest to the T-Bills?
- 4 If your risk aversion is $A = 15$, what should be the weights of your optimal complete portfolio on the efficient frontier generated by A1, A2, A3 and the T-Bills?
- 5 What should be the weight of the complete portfolio in stock A3?